Doctor NiGONiGO's multi-core CPU

Tags:

Time Limit: 3 s Memory Limit: 64 MB
Submission: 5 AC: 2 Score: 99.85

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Description

Doctor NiGONiGO has developed a new kind of multi-core CPU, it works as follow. There are q ($1 < q \le 50$) identical cores in the CPU, and there are p ($0 < p \le 200$) jobs need to be done. Each job should run in some core and need only one unit time. The cores in the CPU can work simultaneously but each core can implement only one job at a time. When a job completed, it will incur a deferral cost. Job i has a deferral cost $c(i, j)$ ($0 \le c(i, j) \le 1000000$), where j is the completion time of the job ($1 \le j \le p$ because any job done in the time later than p obviously not the optimal solution). Here we assume that $c(i, j)$ is a monotonically no decreasing function of j. Now you are asked to find the schedule which have the minimum overall deferral cost.

Input

An integer T indicated the number of test cases.

For each test case:

There are two integers q, p in first line.

And Following is p lines, each line contain p integer, the jth integer of the ith line is the deferral cost $c(i, j)$.

Output

For each test case, output a singer integer, which is the minimum total deferral cost of the Problem.

Samples

input Copy

```
1
2 4
89 145 181 269
4 86 158 164
60 143 157 165
4 45 109 207
```

output Copy

```
254
```

Hint

a. In the sample case, core 1 execute the job 2, 1 in sequence; core 2 execute the job 3, 4 in sequence, thus the optimal solution has the overall cost (4 + 145) + (60 + 45) = 254.
b. Huge input.

Source

3rd Central South China Programming Contest

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